Guillaume ASTRUC

PhD Student

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Education

2025–2025 **Research Student**, *Google*, Zürich.
Subject: Unified Geospatial Contrastive Learning
Advisors: Paul-Edouard Sarlin, Eduard Trulls and Jan Hosang

2023–present **PhD, Computer Science & Engineering**, *ENPC/IGN*, Paris.
Subject: Self-supervised cross-modal geospatial learning
Advisors: Loic Landrieu, Nicolas Gonthier and Clément Mallet

2021–2022 : **Master Data Science**, *Institut Polytechnique de Paris*, Palaiseau, France.

2018–2022 : **Engineering school**, *ENSAE Paris*, Palaiseau, France, Graduated statistician-economist engineer.

Academic Activities

Publications

2026 **Yohann Perron, Guillaume Astruc, Nicolas Gonthier, Clement Mallet, Loic Landrieu**, UniverSat: Resolution-and Modality-Agnostic Transformers for Earth Observation, *NeurIPS 2026*.

2026 **Guillaume Astruc, Eduard Trulls, Jan Hosang, Loic Landrieu, Paul-Edouard Sarlin**, UNIGEOCLIP: Unified Geospatial Contrastive Learning, *EarthVision (CVPRW) 2026*.

2024 **Guillaume Astruc, Nicolas Gonthier, Clement Mallet, Loic Landrieu**, AnySat: An Earth Observation Model for Any Resolutions, Scales, and Modalities, *CVPR 2025*.
Highlight (Top 10%)

2024 **Guillaume Astruc, Nicolas Gonthier, Clement Mallet, Loic Landrieu**, OmniSat: Self-Supervised Modality Fusion for Earth Observation, *ECCV 2024*.

2024 **Guillaume Astruc, Nicolas Dufour, Ioannis Siglidis, Constantin Aronssohn, Nacim Bouia, Stephanie Fu, Romain Loiseau, Van Nguyen Nguyen, Charles Raude, Elliot Vincent, Lintao XU, Hongyu Zhou, Loic Landrieu**, OpenStreetView-5M: The Many Roads to Global Visual Geolocation, *CVPR 2024*.

Teaching

2025 - 2026 **Teacher**, *Deep Learning for Remote Sensing*, ENSG.
- Deep Learning for 3D
- Deep Learning for Temporal Data

2025 **Teaching Assistant**, *Introduction to Data Science*, ENPC.
- Statistical Inference
- Linear and Logistic Regression Models
- PCA
- Clustering
- Validation and Regularization
- Introduction to Deep Learning

2024-2026 **Teaching Assistant**, *Deep Learning for Remote Sensing*, ENSG.
- Intro to Deep Learning
- Deep Learning for Multispectral Images
- Deep Learning for 3D Data
- Deep Learning for Temporal Data

2024 **Teaching Assistant**, *Machine Learning*, ENSG.
- Supervised Learning
- Classification

Reviewing

2027 **Reviewer for ICLR**.
2026 **Reviewer for ECCV**.
2026 **Reviewer for CVPR**, *Oustanding Reviewer Award, 2026*.
2025 - 2026 **Reviewer for NeurIPS**.
2024 - 2026 **Reviewer for EarthVision (CVPRW)**.

Recent open-source repositories

97 ★, 7 ⑂ **gastruc/UniverSat**, *Owner*.
21 ★, 5 ⑂ **google-research/scaling-geoloc**, *Contributor*.
46 ★, 1 ⑂ **gastruc/UniGeoClip**, *Owner*.
218 ★, 23 ⑂ **gastruc/AnySat**, *Owner*.
99 ★, 8 ⑂ **gastruc/OmniSat**, *Owner*.
189 ★, 15 ⑂ **gastruc/osv5m**, *Owner*.

Research Experience

2025 – 6 months ***Unified Geospatial Contrastive Learning***, *Google*, Zürich, Switzerland.
Advisor : **Dr. Paul-Edouard Sarlin**, *Research Scientist*, Zürich, Switzerland (*Personal Webpage*)

2022 – 6 months ***Unsupervised Re-Identification for child-clinician recognition for Autism behaviour detection***, *INRIA*, Sophia-Antipolis, France.
Advisor : **Dr. François Bremond**, *Research Director*, Sophia-Antipolis, France (*Personal Webpage*)

2021 – 6 months ***"Deep reinforcement learning and deep learning 3D point cloud classification"***, *ONERA*, Palaiseau, France.
Advisor : **Dr. Adrien Chan Hon Tong**, *Researcher*, Palaiseau, France (*Personal Webpage*)

2020 – 6 months ***"Multi-temporal data and segmentation from satellite imagery to create a macroeconomic indicator"***, *QuantCube Technology*, Paris, France.

2020 – 3 months ***"Robust estimation of the mean by a Median-of-Means approach and Tukey depth"***, *CREST*, Palaiseau, France.
Advisor : **Prof. Guillaume Lecué**, *Professor*, Cergy, France (*Personal Webpage*)

Miscellaneous

Research Interests

- Computer Vision
- Self-supervised learning
- Cross-modal learning
- Remote Sensing
- Foundation models

Languages spoken

French, *Native*
English, *Fluent*, C1
German, B2

Associative Commitments

2024-2026 Member of the PUC Running Bureau
2019-2020 Vice-President of the ENSAE Student Office
2019-2020 Member of the ENSAE Sports Bureau